

The Quantum Structural Triad: Fluctuations, Entropy, and Correlations as Interdependent Primitives

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Abstract

We identify a primitive structural condition in quantum theory: energy fluctuations, von Neumann entropy, and quantum mutual information constitute an inseparable triad that cannot vanish on average within any physically realizable quantum system exhibiting local coupling. This triad expresses the minimal structural integrity required for a quantum state to persist, evolve, and mediate phenomena. From the assumptions of locality and dynamical non-stationarity we derive two complementary inequalities—a Hamiltonian-capped and a speed-based form—that establish explicit quantitative thresholds for systemic viability. These thresholds reveal how fluctuations, entropy, and correlations must coexist and compensate one another to sustain quantum activity. The results apply to pure and mixed states, extend naturally to open systems via entropy production, and generalize to algebraic QFT. Beyond theory, the framework provides operational criteria for assessing structural viability in quantum technologies, from qubit certification to memory stability and error-correction design. While subsystem quantities are used for analytical resolution, the theory is fundamentally systemic: it diagnoses the coherence and persistence of the total quantum state rather than the autonomy of its parts. We further conjecture that analogous structural constraints may extend to globally topological quantum systems, whose persistent entropic and correlational structure may reflect a generalized form of systemic viability.

1 Introduction

Quantum theory imposes fundamental constraints on what qualifies as a physically viable subsystem—understood here as a resolvable region within an interacting system. From the uncertainty principle to modern information-theoretic bounds[2, 7], it has long been clear that fluctuations, entropy, and correlations cannot all vanish simultaneously without undermining the internal consistency of the theory. Each captures a distinct aspect of quantum structure: energy fluctuations quantify sensitivity to local generators[1, 21], von Neumann entropy measures local information content[5, 19], and mutual information encodes connectivity across subsystems[8, 10]. While each has been widely studied in isolation or in pairs— as resources for metrology[3, 22], thermodynamics[20, 11], or information processing[12]— their collective interdependence has remained uncharacterized.

Existing work has established important pairwise constraints: generalized speed limits connect quantum Fisher information to entropy production[4, 27], fluctuation–speed relations bound entropy growth in open systems[24, 25], and entanglement persists under noise through correlation–entropy trade-offs[29, 28]. Yet none of these results capture the essential *three-way structural interdependence* that we show to be indispensable for system viability.

Our central claim is that fluctuation, entropy, and correlation form a *quantum structural triad* that in any physically realizable quantum system exhibiting local coupling, their product cannot vanish.

Pointwise vanishing may occur, but systemic persistence is mandatory. This defines a minimal structural threshold below which a subsystem becomes *quantum-structurally inert*[22, 26]: incapable of sustaining dynamics, storing information, or mediating interactions. In contrast, persistent nonvanishing of the triad guarantees that a subsystem possesses the minimal structural integrity required to remain quantum-mechanically relevant.

The novelty lies not in recognizing that these three quantities matter, but in deriving explicit, unit-correct *quantitative thresholds* that determine how much of each is required and how they trade off against

one another. These thresholds scale nontrivially with system size and minimal energy scales[18, 16], providing a falsifiable criterion for subsystem viability. Unlike heuristic measures of coherence, our results emerge from first principles, yielding both theoretical universality and practical tools for quantum technologies.

Problem statement. Consider a system on a bounded-degree finite-range interaction graph and any bipartition $(A|\bar{A})$ crossed by at least one interaction. Define the triad $\Pi_A(t) = \sqrt{F_Q(\rho_A; H_A)} S_A I(A:\bar{A})$. Without presupposing positivity of any leg, establish necessary non-inertness: prove that the triad cannot vanish in time-average across the cut-set under locality, genuine cross-cut coupling, and global non-stationarity. [6, 2, 1]

2 Quantum Structural Triad: Definition and Role

We define the *quantum structural triad* of a subsystem A in a global state $\rho(t)$ as

$$\Pi_A(t) = \sqrt{F_Q(\rho_A(t); H_A)} S_A(t) I(A : \bar{A}, t), \quad (1)$$

where:

- $F_Q(\rho_A; H_A)$ is the quantum Fisher information of the reduced state ρ_A with respect to the local generator H_A , equivalent to the Bures metric curvature at time t , capturing *fluctuation strength*.
- $S_A(t) = -\text{Tr}[\rho_A(t) \log \rho_A(t)]$ is the von Neumann entropy of the subsystem, quantifying *local informational content*.
- $I(A : \bar{A}, t) = S_A(t) + S_{\bar{A}}(t) - S_{A\bar{A}}(t)$ is the quantum mutual information, measuring total (quantum and classical) *correlation* between A and its complement.

Each leg reflects a distinct facet of quantum structure:

1. $\sqrt{F_Q}$ encodes local dynamical sensitivity to perturbations,
2. S_A encodes intrinsic informational richness,
3. $I(A : \bar{A})$ encodes connectivity to the rest of the system.

Interpretation. We interpret $\Pi_A(t)$ as the *instantaneous structural integrity* of subsystem A . If $\Pi_A(t) = 0$, then A is simultaneously fluctuation-free, pure, or decorrelated, and thus fails to contribute meaningfully to the system's quantum structure.

Quantum-Structurally Inert (QSI).

Definition 1 (D2 (QSI on an epoch)). *A subsystem A is quantum-structurally inert on a measurable epoch $K \subset [0, \infty)$ with $\delta(K) > 0$ if $\Pi_A(t) = 0$ for almost every $t \in K$.*

Systemic Triad. We define the systemic measure as the sum over all subsystems:

$$\Pi(t) = \sum_A \Pi_A(t). \quad (2)$$

Our main result is that $\Pi(t)$, while possibly vanishing at isolated instants, cannot vanish on average across subsystems and over time in any locally coupled, physically viable quantum system. This establishes the triad as a *minimal structural threshold for subsystem viability*.

3 Hypotheses and Objects

Conventions. All logarithms are natural; entropies are measured in nats. For any subsystem A , let d_A denote the Hilbert space dimension of H_A and $d_{\bar{A}}$ that of its complement.

Interaction graph and cut-set. We represent the system by a bounded-degree, finite-range interaction graph $G = (V, E)$ associated with the local terms of the Hamiltonian. A cut-set E consists of the nearest-neighbor edges crossing a bipartition $(A|\bar{A})$. Its cardinality is denoted $|E|$, and for intensive averages we use

$$\frac{1}{|E|} \sum_{A \in E} (\cdots). \quad (3)$$

Structural Hypotheses (H1–H3). To state our results precisely, we impose three physically natural hypotheses:

H1. Locality. The Hamiltonian is a sum of local terms with finite interaction range and uniformly bounded degree on the interaction graph G .

H2. Genuine coupling. Each bipartition defined by the cut-set E includes at least one nonzero coupling term, preventing complete isolation of subsystems.

H3. Global non-stationarity. There exists a measurable set $K \subset [0, \infty)$ with lower density $\delta > 0$ such that $\dot{\rho}(t) \neq 0$ for almost every $t \in K$ (closed systems), or such that the CPTP evolution Λ_t is non-stationary on K (open systems).

Continuity caveat. Rank-change times can accumulate in engineered cases. Our results apply on *full-rank epochs*, where $\sqrt{F_Q}$, S_A , and $I(A : \bar{A})$ are continuous (see Lemma 3).

Dimension bounds. For any bipartition $(A, \bar{A})$, the mutual information is bounded by:

$$I(A : \bar{A}) \leq 2 \log \min(d_A, d_{\bar{A}}). \quad (4)$$

Interpretation. These hypotheses exclude trivial or degenerate regimes (e.g., completely isolated subsystems, frozen dynamics, or rank-changing discontinuities). Within this admissible domain, we establish that the triad Π_A defines a nonvanishing *structural threshold* for subsystem viability.

Scope Note. Although expressed through subsystem observables, the framework is not subsystem-reductionist. The triad thresholds are intended to characterize the *global* viability of the quantum state, with partitions introduced as analytical tools for structural resolution. We make no ontological claim about subsystems as independently real; they serve as a lens through which system-level structure becomes operationally visible.

4 Geometric Uncertainty: QSL from Bures Angle

The Mandelstam–Tamm form of the quantum speed limit provides a direct link between distinguishability in state space and local energy fluctuations. For a subsystem A , the minimal time τ_A required to evolve by a Bures angle L obeys

$$\tau_A \geq \frac{\hbar L}{\Delta E_A}, \quad (5)$$

where ΔE_A is the time-averaged standard deviation of the local Hamiltonian H_A in the state ρ_A .

Equivalently,

$$\Delta E_A \geq \frac{\hbar L}{\tau_A}. \quad (6)$$

Interpretation. This expresses the *geometric face of uncertainty*: whenever the subsystem undergoes a “distinguishable change” (nonzero Bures angle), local energy fluctuations must be strictly positive. In structural terms, no subsystem can remain indefinitely fluctuation-free while still evolving in Hilbert space. This provides the first leg of the structural triad.

5 Entropic Response: First Law of Entanglement

For small perturbations of a quantum state, the first law of entanglement entropy provides a direct link between entropy variation and modular Hamiltonian expectation values:

$$\delta S_A = \delta \langle H_A^{\text{mod}} \rangle, \quad (7)$$

where H_A^{mod} is the modular Hamiltonian associated with region A .

Interpretation. This relation shows that entropy cannot remain frozen under nontrivial local energy exchange. Whenever the subsystem participates in dynamics, changes in S_A are unavoidable. Entropy thus forms the second leg of the structural triad, encoding the local informational footprint of the subsystem.

Caveat. The first law of entanglement is exact for small variations around a reference state, and is most transparent when modular Hamiltonians are explicitly known (e.g., conformal field theory or holographic settings). For general mixed states, it should be interpreted as a local differential relation rather than a global identity. Nevertheless, it captures the essential fact: local entropy cannot vanish persistently in a coupled, evolving quantum system.

6 Correlation as Intrinsic

The quantum mutual information across a bipartition $(A|\bar{A})$ is defined as

$$I(A : \bar{A}) = S_A + S_{\bar{A}} - S_{A\bar{A}}. \quad (8)$$

It measures the total correlation—both classical and quantum—between the subsystem A and its complement $\bar{A}$.

Interpretation. Entropy is never a purely local quantity: it necessarily reflects correlations with the environment or complementary subsystems. Thus mutual information, as the third leg of the triad, is structurally unavoidable in any nontrivial quantum regime. If correlations vanished identically across a genuine cut, the subsystem would be decoupled and structurally irrelevant.

Structural role. Correlation ensures that local fluctuations and entropy are not isolated artifacts but are embedded in a larger relational structure. Together with $\sqrt{F_Q}$ and S_A , mutual information completes the minimal triad of structural viability.

7 Per-Subsystem Inequality

Combining the fluctuation, entropy, and correlation terms, and using the instantaneous Bures speed bound, we obtain on any active interval:

$$\Delta E_A S_A I(A : \bar{A}) \geq \frac{\hbar}{2} \dot{L}_B S_A I(A : \bar{A}), \quad (9)$$

where $\dot{L}_B$ is the instantaneous Bures angle rate.

Integrating over time and applying the density-to-time-average lemma on a measurable active set K with lower density $\delta > 0$, we arrive at

$$\langle \Pi_A \rangle_t \geq \delta L \frac{S_A I(A : \bar{A})}{\tau_A}. \quad (10)$$

Interpretation. This inequality expresses the *structural inseparability* of the triad. Variance, entropy, and correlation cannot collapse simultaneously for a genuinely coupled subsystem. At least one leg must sustain a strictly positive floor over active epochs, ensuring that Π_A cannot vanish in time average.

Structural consequence. Whenever $\langle \Pi_A \rangle_t = 0$, the subsystem is quantum-structurally inert: it contributes no persistent fluctuations, entropy, or correlations. The inequality therefore defines a *per-subsystem structural threshold for viability*.

8 Systemic Inequality

Summing over all subsystems yields the global triad inequality:

$$\Pi(t) = \sum_A \sqrt{F_Q(\rho_A(t); H_A)} S_A(t) I(A : \bar{A}, t) \geq \sum_A \frac{L S_A I(A : \bar{A})}{\tau_A}. \quad (11)$$

As long as the trajectory in state space has nonzero Bures length ($L > 0$), the right-hand side is strictly positive.

Interpretation. Whenever the global state undergoes distinguishable evolution, the triad enforces a *system-wide structural threshold*. Even if individual subsystems fluctuate between relevance and irrelevance at specific instants, the ensemble cannot sustain a collectively vanishing product across time.

Structural consequence. This inequality establishes that no physically viable, locally coupled quantum system can remain entirely fluctuationless, unentangled, and pure across all subsystems. Systemic persistence of the triad is thus a universal requirement for maintaining quantum structural integrity.

9 Intensive Formulation and Root Inequalities

9.1 Normalized Intensive Triad

To compare across systems and isolate structure from scale, we define a dimensionless intensive version of the triad for each cut $(A|\bar{A})$:

$$\hat{\Psi}_A(t) := \underbrace{\frac{\hbar \sqrt{F_Q(\rho_A(t); H_A)}}{2 \|H_A\|_{\text{op}}}}_{\in [0,1]} \cdot \underbrace{\frac{S_A(t)}{\log d_A}}_{\in [0,1]} \cdot \underbrace{\frac{I(A : \bar{A}, t)}{2 \log \min(d_A, d_{\bar{A}})}}_{\in [0,1]}. \quad (12)$$

Each factor is bounded by standard inequalities, so $0 \leq \hat{\Psi}_A(t) \leq 1$. We use the cut-set average and its time average:

$$\hat{\Psi}_{\text{cut}}(t) := \frac{1}{|E|} \sum_{A \in E} \hat{\Psi}_A(t), \quad \langle \hat{\Psi}_{\text{cut}} \rangle_t := \liminf_{T \rightarrow \infty} \frac{1}{T} \int_0^T \hat{\Psi}_{\text{cut}}(t) dt.$$

9.2 Hamiltonian-Capped Structural Threshold

Theorem 1 (Hamiltonian-Capped Structural Threshold). *Under H1–H2 (no speed assumptions), for any cut-set E ,*

$$\frac{1}{|E|} \sum_{A \in E} \Delta E_A S_A I(A : \bar{A}) \geq 2 (\log d_{\min})^2 \|H\|_{\min} \langle \hat{\Psi}_{\text{cut}} \rangle_t, \quad d_{\min} := \min_{A \in E} \min(d_A, d_{\bar{A}}), \quad (13)$$

where $\|H\|_{\min} := \min_{A \in E} \|H_A\|_{\text{op}}$.

Proof sketch. From (12),

$$\sqrt{F_Q} S_A I(A : \bar{A}) := \frac{2 \|H_A\|_{\text{op}}}{\hbar} (\log d_A) [2 \log \min(d_A, d_{\bar{A}})] \hat{\Psi}_A(t).$$

Use $\Delta E_A \geq \frac{\hbar}{2} \sqrt{F_Q}$ to replace $\sqrt{F_Q}$, then average over $A \in E$ and lower-bound $\|H_A\|_{\text{op}}$, $\log d_A$, $\log \min(d_A, d_{\bar{A}})$ by their respective minima. Time-averaging yields (13). $\square$

Interpretation. This threshold is *unconditional in speed*: even without assuming a dynamical floor, the normalized structural integrity $\langle \hat{\Psi}_{\text{cut}} \rangle_t$ forces a positive lower bound set by the *minimal local energy scale* and *minimal subsystem dimension*. It quantifies how structural integrity trades off against hardware scales.

9.3 Speed-Based Structural Threshold

Theorem 2 (Speed-Based Structural Threshold). *Assume H1–H2–H3. By Lemmas L2–L4, there exist $\varepsilon_F, \varepsilon_S, \varepsilon_I > 0$ and a measurable subset $K^* \subset K$ with $\delta(K^*) > 0$ such that*

$$\sqrt{F_Q(\rho_A(t); H_A)} \geq \varepsilon_F, \quad S_A(t) \geq \varepsilon_S, \quad I(A : \bar{A}, t) \geq \varepsilon_I$$

for almost every $t \in K^*$ and all $A \in E$. Then

$$\frac{1}{|E|} \sum_{A \in E} \Delta E_A S_A I(A : \bar{A}) \geq \frac{\hbar}{2} \delta(K^*) \varepsilon_F \varepsilon_S \varepsilon_I. \quad (14)$$

This uses the density-to-time-average lemma.

Interpretation. This threshold is *dynamical*: it converts a Bures/QFI speed floor on a derived active epoch $K^* \subset K$ into a strictly positive structural lower bound, with explicit constants $(\hbar/2) \delta(K^*) \varepsilon_F \varepsilon_S \varepsilon_I$. It is typically tighter during nonequilibrium evolution. Finite intersections of positive-measure sets retain positive measure; density-to-time-average then yields the stated bound.

9.4 Trade-offs and Scaling

Both thresholds quantify *trade-offs*: if one leg (fluctuation, entropy, or correlation) is small, the others must compensate to keep the left-hand side above threshold. System-size enters via $(\log d_{\min})^2$: larger minimal Hilbert spaces demand stronger combined resources to maintain structural integrity.

10 Technical Lemmas

The following lemmas establish the auxiliary bounds needed to derive the structural thresholds.

Lemma 1 (QFI-variance bound). *For any density operator ρ and Hermitian generator H ,*

$$F_Q(\rho; H) \leq \frac{4}{\hbar^2} \text{Var}_\rho(H), \quad \Delta E \geq \frac{\hbar}{2} \sqrt{F_Q(\rho; H)}. \quad (15)$$

Lemma 2 (Instantaneous Bures speed). *For unitary dynamics generated by H_A ,*

$$\dot{L}_B(t) = \frac{1}{2} \sqrt{F_Q(\rho_A(t); H_A)}. \quad (16)$$

For general CPTP dynamics, $\dot{L}_B(t) \leq \frac{1}{2} \sqrt{F_Q}$.

Lemma 3 (Continuity and full-rank epochs). *For finite-dimensional piecewise C^1 trajectories, the functions $t \mapsto S_A(t)$, $t \mapsto I(A : \bar{A}, t)$, and $t \mapsto \sqrt{F_Q(\rho_A(t); H_A)}$ are continuous on intervals where $\rho_A(t)$ is full rank. Rank-change times form a set of measure zero.*

Definition 2 (D1 (Lower density)). *For measurable $K \subseteq [0, \infty)$, define*

$$\delta(K) := \liminf_{T \rightarrow \infty} \frac{|K \cap [0, T]|}{T}. \quad (17)$$

Lemma 4 (Density-to-time-average). *Let $f : [0, \infty) \rightarrow [0, \infty)$ be measurable, and suppose $f(t) \geq \alpha > 0$ for all $t \in K$ with $\delta(K) > 0$. Then*

$$\liminf_{T \rightarrow \infty} \frac{1}{T} \int_0^T f(t) dt \geq \delta(K) \alpha. \quad (18)$$

Structural Lemmas (L1–L4).

Lemma 5 (L1: Reduced motion from global motion). *Under H1–H2–H3, reduced dynamics is non-stationary for almost every time in K except possibly a null set: fine-tuned decoupling across a genuine cut has measure zero.*

Lemma 6 (L2: Fluctuation leg). *From non-stationarity on K and Lemma 2, there exists a subset $K_F \subset K$ with $\delta(K_F) > 0$ and $\sqrt{F_Q(\rho_A; H_A)} > 0$ for almost every $t \in K_F$.*

Lemma 7 (L3: Correlation leg). *With a cross-cut coupling (H2) and locality (H1), Lieb–Robinson propagation implies $I(A : \bar{A}) > 0$ after a microscopic time on a positive-measure subset $K_I \subset K$.*

Lemma 8 (L4: Entropy leg). *For closed pure global states, $I = 2S_A$. For mixed/open dynamics, Spohn’s inequality implies strictly positive local entropy production away from steady state on a positive-measure subset $K_S \subset K$.*

11 Discussion: Structural Relevance and Irrelevance

11.1 Complementary Thresholds

The two inequalities derived above provide complementary perspectives on subsystem viability:

- **Hamiltonian-capped threshold:** unconditional, tied to the smallest operator norm $\|H\|_{\min}$ and subsystem dimension $d_{\min}$. It reflects the *structural potential* inherent in the system’s composition.
- **Speed-based threshold:** conditional on a nonzero Bures/QFI speed floor, it converts dynamical activity into a strictly positive structural lower bound, typically tighter in nonequilibrium regimes.

These represent the only two physically consistent, unit-correct routes to nonvanishing lower bounds on the fluctuation–entropy–correlation product.

11.2 Units and Physical Meaning

The structural triad combines distinct units:

- $\sqrt{F_Q}$ carries units of inverse time (distinguishability rate),
- ΔE carries energy,
- S and I are dimensionless (nats).

The Hamiltonian-capped threshold introduces an energy scale via $\|H\|_{\min}$, while the speed-based threshold introduces a temporal scale via the Bures speed. In both cases, the inequality enforces that the product of these resources cannot vanish under admissible conditions.

11.3 Boundary Cases and Structural Irrelevance

The thresholds become trivial precisely when an *excluded hypothesis* holds:

1. $I \equiv 0$ (violates H2: no genuine cross-cut coupling),
2. $\sqrt{F_Q} \equiv 0$ or zero-speed protocols (violates H3: global non-stationarity),
3. Rank-changing discontinuities beyond measure-zero sets (covered by continuity caveat).

In each case, the subsystem is *quantum-structurally inert* (QSI): incapable of acting as a reliable mediator, memory, or participant in nontrivial dynamics.

11.4 Trade-offs and Design Rules

The inequalities highlight explicit trade-offs:

- Small entropy can be compensated by larger fluctuations or correlations,
- Weak correlations demand stronger fluctuations and entropy,
- Minimal subsystem dimension $d_{\min}$ sets a rising structural cost with scale.

Thus subsystem viability is governed not by any single resource but by their interdependence. This suggests practical design rules: viable subsystems must maintain all three legs above thresholds in some combination, and resource deficits in one leg require reinforcement in the others.

11.5 Experimental Accessibility

Few-body experiments can probe the triad directly via:

- QFI estimates from Bures angle trajectories,
- Entropy via tomography or entropic witnesses,
- Mutual information via correlation measurements.

Many-body systems require witness-based or coarse estimators, but our finite-window QFI bound (Appendix) enables positive certification even with partial data.

Corollary (no admissible QSI epochs)

If H1–H2–H3 hold, then A cannot be QSI on K : otherwise $\Pi_A = 0$ almost everywhere on a positive-density epoch K^* contradicts the thresholds above.

12 Quantum Computing Applications

The structural thresholds provide practical, physics-based exclusion criteria for quantum technologies. Because they are *necessary conditions*, any subsystem falling below threshold is *guaranteed* to lack quantum functionality, allowing definitive elimination of nonviable qubits, gates, or memories.

12.1 Qubit Viability Certification

For each physical qubit i , measure or estimate

$$\Pi_i = \sqrt{F_Q(\rho_i; H_i)} S_i I(i : \bar{i}). \quad (19)$$

A qubit is structurally viable only if

$$\Pi_i \geq \max\{\text{Hamiltonian-capped threshold, Speed-based threshold}\}.$$

Qubits with Π_i below threshold are quantum-structurally inert and cannot support reliable quantum logic.

12.2 Error Correction Prioritization

Error correction is resource-intensive. Our framework identifies qubits worth correcting:

$$\text{Apply QEC only if } \Pi_i \geq \text{threshold}.$$

Qubits below threshold are structurally irrelevant and cannot be salvaged by error correction, saving resources in large-scale devices.

12.3 Quantum Memory Lifetime

The speed-based inequality provides a bound on memory coherence:

$$\text{Quantum memory is viable while } \delta(t) \varepsilon_F(t) \varepsilon_S(t) \varepsilon_I(t) \geq \frac{2}{\hbar} \Pi_{\min}. \quad (20)$$

When any component falls below its floor, the memory is structurally incapable of preserving quantum coherence.

12.4 Gate Operation Certification

For multi-qubit gates, structural viability requires

$$\Pi_{\text{gate}} \geq \frac{\hbar}{2} \delta_{\text{gate}} \varepsilon_F \varepsilon_S \varepsilon_I \quad (21)$$

throughout the gate execution. Gates failing this criterion cannot realize genuine quantum logic, regardless of classical fidelity measures.

12.5 NISQ Device Optimization

In noisy intermediate-scale devices, the thresholds enable:

- identification of qubits that are “quantum enough” to include in algorithms,
- bounding of maximum circuit depth before triad decay below threshold,
- optimization of connectivity patterns based on correlation persistence.

12.6 Towards Principled Design

Current practice in quantum engineering is heuristic, guided by coherence times and error rates. Our framework introduces principled design rules grounded in first-principles physics. It answers questions such as:

- “Is this qubit structurally capable of quantum computation?”
- “When should error correction be abandoned?”
- “What are the fundamental limits of this processor architecture?”

By providing definitive exclusion criteria, the structural triad transforms quantum engineering from empirical tuning to threshold-driven design.

13 Open-System Extension: Structural Relevance Under Dissipation

13.1 Setting and Hypotheses

We extend the structural triad to dissipative dynamics governed by a local Lindbladian

$$\mathcal{L} = \sum_X \mathcal{L}_X,$$

with finite interaction range and bounded degree. The CPTP propagator is $\Lambda_t = e^{t\mathcal{L}}$ (or time-periodic for driven steady states). Cuts are defined by the same nearest-neighbor cut-set E . Each cut $(A|\bar{A})$ is assumed genuinely coupled by at least one Lindblad or Hamiltonian term. Dissipative Lieb–Robinson bounds ensure finite-speed propagation of influence.

H3 (open). There exists a measurable epoch $K \subset [0, \infty)$ of non-stationary CPTP evolution (away from a fixed point) with $\delta(K) > 0$.

13.2 Open-System Triad Reformulation

Even under dissipation, the triadic structure persists in generalized form:

- **Fluctuations:** The QFI leg is replaced by a channel speed measure F_{chan} , defined via monotone Riemannian metrics on CPTP maps.
- **Entropy:** The entropy leg becomes the local entropy production rate

$$\dot{S}_{\text{prod},A}(t) := -\frac{d}{dt} D(\rho_A(t) \| \sigma_A) \geq 0,$$

where σ_A is the full-rank local steady state (Spohn's inequality), strictly > 0 away from steady state.

- **Correlations:** The correlation leg is extended to system–environment mutual information, accessible via Stinespring dilations.

13.3 Intensive Open-System Triad

Define the normalized intensive open-system triad:

$$\hat{\Psi}_A^{\text{open}}(t) := \underbrace{\frac{\sqrt{F_{\text{chan}}(\Lambda_t|A)}}{\sqrt{F_{\text{max}}}}}_{\in[0,1]} \cdot \underbrace{\frac{\dot{S}_{\text{prod},A}(t)}{\dot{S}_{\text{max}}}}_{\in[0,1]} \cdot \underbrace{\frac{I(\text{sys} : \text{rest}, t)}{2 \log d_{\text{min}}}}_{\in[0,1]}. \quad (22)$$

The cut-set intensive average is

$$\hat{\Psi}_{\text{cut}}^{\text{open}}(t) = \frac{1}{|E|} \sum_{A \in E} \hat{\Psi}_A^{\text{open}}(t), \quad \overline{\hat{\Psi}_{\text{cut}}^{\text{open}}} := \liminf_{T \rightarrow \infty} \frac{1}{T} \int_0^T \hat{\Psi}_{\text{cut}}^{\text{open}}(t) dt.$$

Theorem 3 (Open-System Structural Threshold). *Under locality, genuine coupling, finite-speed dissipative propagation, and existence of a non-stationary epoch K with density $\delta > 0$ and leg-wise positivity away from steady state,*

$$\overline{\hat{\Psi}_{\text{cut}}^{\text{open}}} \geq \delta \frac{\varepsilon_F}{\sqrt{F_{\text{max}}}} \cdot \frac{\varepsilon_S}{\dot{S}_{\text{max}}} \cdot \frac{\varepsilon_I}{2 \log d_{\text{min}}} > 0. \quad (23)$$

Interpretation. Even with dissipation, subsystems cannot persist in a state of structural irrelevance except at true fixed points. Any nontrivial open-system evolution enforces persistent nonvanishing of fluctuation, entropy production, and correlations, maintaining quantum structural integrity in the presence of noise and dissipation.

14 Conjecture: Topological Structural Triad

The structural triad, as derived above, presupposes locality and dynamical non-stationarity (H1–H3). However, certain quantum systems—notably the toric code—exhibit structural persistence in the absence of both. This motivates a conjectural extension of the framework to globally topological, non-dynamical quantum orders.

Mathematical Structure

The toric code Hamiltonian on an $L \times L$ torus,

$$H = -J_e \sum_v A_v - J_m \sum_p B_p, \quad (24)$$

acts on $N = 2L^2$ qubits located on the edges of the lattice. Here $A_v = \prod_{i \in v} \sigma_i^x$ and $B_p = \prod_{i \in p} \sigma_i^z$ are the mutually commuting star and plaquette stabilizers. Despite its static ground-space structure, the toric code retains finite entropy, mutual information, and logical fluctuations under nonlocal observables.

Ground-space structure. All stabilizers commute and satisfy $A_v^2 = B_p^2 = I$, yielding

$$\dim \mathcal{H}_{\text{GS}} = 4, \quad (25)$$

with logical operators W_x, W_z forming noncontractible Wilson loops that anticommute on the torus.

Logical fluctuations. For the mixed ground-state density operator,

$$\rho_{\text{mix}} = \frac{1}{4} \sum_{i=1}^4 |\psi_i\rangle\langle\psi_i|, \quad (26)$$

the logical Wilson loops satisfy

$$\langle W_x \rangle = \langle W_z \rangle = 0, \quad \Delta W_x = \Delta W_z = 1, \quad (27)$$

indicating persistent logical uncertainty even in the absence of dynamical evolution.

Entropic and correlational structure. For half-torus bipartitions at $L = 2$,

$$S_A \approx 2.77 \text{ nats}, \quad (28)$$

$$I(A : \bar{A}) \approx 4.16, \quad (29)$$

leading to the observed topological triad values

$$\Pi_{x,z}^{\text{topo}} = \Delta W_{x,z} S_A I(A : \bar{A}) \approx 11.53. \quad (30)$$

The equality $\Pi_x^{\text{topo}} = \Pi_z^{\text{topo}}$ reflects isotropy under $J_e = J_m$.

Interpretation

Even without dynamical evolution or local coupling, the triad remains finite. The logical fluctuations $\Delta W_{x,z}$ substitute for local energy variance, while S_A and $I(A : \bar{A})$ preserve their informational roles. The toric code therefore satisfies a generalized static inequality,

$$\Pi^{\text{topo}} = \Delta_{\text{logical}} S_A I(A : \bar{A}) > 0, \quad (31)$$

expressing the structural viability of topological quantum systems.

Implications

- The persistence of $\Pi^{\text{topo}} > 0$ demonstrates that systemic viability can be maintained through global logical structure, not only through dynamical coupling.
- Logical operators act as “virtual dynamics,” sustaining distinguishability and informational curvature in a stationary code space.
- This extension suggests that the triad characterizes *systemic structural persistence*—the capacity of a quantum system to remain informationally active—whether sustained by local motion or by topological coherence.

Future Computations

To formalize this conjecture, subsequent work will:

- Extend simulations to $L = 3, 4$ using stabilizer algebra rather than dense diagonalization;
- Evaluate the scaling of S_A , $I(A : \bar{A})$, and Π^{topo} with system size L ;
- Test whether Π^{topo}/L^2 approaches a constant, defining a topological invariant of structural integrity.

This conjectural extension positions the triad as a universal descriptor of quantum structural persistence, valid across both local and topological phases.

15 Conclusions

We have identified a primitive structural condition in quantum theory: the inseparable triad of fluctuations, entropy, and correlations defines the minimal requirement for subsystem viability. No physically realizable, locally coupled quantum system can maintain dynamical relevance while any leg of the triad vanishes on average. The triad thus constitutes the fundamental expression of quantum structural integrity—the capacity to persist, evolve, and mediate phenomena.

From locality and dynamical non-stationarity we derived two complementary quantitative thresholds:

- a **Hamiltonian-capped** inequality, governed by minimal local energy scales and subsystem dimensions,
- a **speed-based** inequality, governed by Bures evolution rates and entropic/correlation floors.

Together, these bounds reveal that fluctuations, entropy, and correlations are structurally interdependent. Any persistent deficit in one component must be balanced by the others, while the simultaneous vanishing of all three marks structural inertness. This inseparability defines the threshold between inert mathematical possibility and physically participatory reality.

Although the framework is expressed through subsystem observables—reduced fluctuations, local entropy, and mutual information—it is not reductionist. Subsystems function only as analytical constructs introduced to resolve structure within the total system. The triad theory is fundamentally systemic: it characterizes the integrity of the full quantum state, employing partitions as diagnostic tools rather than ontological divisions.

Subsystems falling below the derived thresholds are structurally inert: incapable of storing information, mediating interactions, or sustaining distinguishable evolution. In contrast, persistent triadic activity guarantees the minimal conditions for quantum persistence. This exclusion principle applies to pure and mixed states, extends naturally to Lindbladian open systems through entropy production, and generalizes to algebraic QFT.

Beyond foundational implications, the triad provides first-principles benchmarks for quantum technologies—criteria for qubit viability, memory stability, gate certification, and error-correction prioritization. It redefines quantum engineering in terms of structural necessity rather than empirical performance.

Future work will extend the computational annexes, formalize experimental witness protocols, and explore the link between subsystem structure, modular dynamics, and emergent geometry. Through this lens, quantum viability is not an operational convention but a universal principle of physical structure.

Acknowledgements

This project began long time ago, with a long-standing question: how does the principle *natura abhorret vacuum*—the idea that physical reality never admits absolute emptiness—retain its validity from the microscopic to the cosmological scale? To explore that question I carried out a symbolic reduction of representative equations drawn from quantum, thermodynamic, relativistic, and cosmological theories. When all non-universal or model-specific quantities were removed, three elements remained invariant across those domains: energy fluctuations, entropy, and Planck’s constant. In pure states, entropy manifests as entanglement entropy, revealing the same structure in its quantum form. These invariants became the foundation of the structural triad developed and verified in this work.

Large-language-model systems were employed as analytical and computational assistants. To limit both human and algorithmic bias and to control potential hallucination, all theoretical results were subsequently verified with the *Rocq* proof assistant and cross-validated through independent numerical simulations.

The full verification materials, including Rocq proof scripts, computational notebooks, and simulation outputs, are available from the author’s GitHub repository upon request.

Although not a professional physicist or formally affiliated researcher, I carried out the conceptual development, interpretation, and synthesis independently. The involvement of AI and proof-assistant tools should be understood as a form of assisted reasoning, not as authorship. I retain complete epistemic responsibility for the arguments, methods, and conclusions advanced here and accept responsibility for any remaining errors.

Annexes: Computational and Formal Backing

The annexes compile notation and normalization details, auxiliary lemmas and proofs, and the full reproducibility package: computational validations A1–A13 alongside the Rocq/Coq verification. For navigation, see the annex index at Section H.

A Notation, Units, and Normalizations

We recall the subsystem triad (Eq. (1))

$$\Pi_A(t) = \sqrt{F_Q(\rho_A(t); H_A)} S_A(t) I(A:\bar{A}, t),$$

and its intensive normalized form (Eq. (12))

$$\hat{\Psi}_A(t) ::= \frac{\hbar \sqrt{F_Q(\rho_A; H_A)}}{2 \|H_A\|_{\text{op}}} \cdot \frac{S_A}{\log d_A} \cdot \frac{I(A:\bar{A})}{2 \log \min(d_A, d_{\bar{A}})} \in [0, 1].$$

Units. With time encoding, $\sqrt{F_Q}$ has units of inverse time, ΔE has units of energy, while S and I are dimensionless (nats). The Hamiltonian-capped inequality introduces an energy scale via $\|H_A\|_{\text{op}}$, and the speed-based form introduces a temporal scale via Bures/QFI speed.

Normalization bounds. (i) $\hbar \sqrt{F_Q}/(2 \|H_A\|_{\text{op}}) \leq 1$ (unitary time encoding; cf. Lemma 1). (ii) $S_A \leq \log d_A$. (iii) $I(A:\bar{A}) \leq 2 \log \min(d_A, d_{\bar{A}})$. Hence $\hat{\Psi}_A \in [0, 1]$.

B Proof Details for Auxiliary Lemmas

Lemma 9 (QFI-variance bound; Lemma 1). *For any density operator ρ and Hermitian generator H , $F_Q(\rho; H) \leq 4 \text{Var}_\rho(H)/\hbar^2$ and thus $\Delta E \geq (\hbar/2) \sqrt{F_Q}$.*

Proof. Using the SLD representation $F_Q(\rho; H) = \text{Tr}(\rho L^2)$ where $\partial_\theta \rho = (L\rho + \rho L)/2$ for the unitary encoding $\rho(\theta) = e^{-\frac{i}{\hbar} \theta H} \rho e^{\frac{i}{\hbar} \theta H}$, one has $\partial_\theta \rho = -\frac{i}{\hbar} [H, \rho]$. Monotonicity of the Bures metric and Cauchy–Schwarz yield the standard inequality $F_Q \leq 4 \text{Var}_\rho(H)/\hbar^2$ (see [1, 32]). Taking square roots gives $\Delta E \geq (\hbar/2) \sqrt{F_Q}$. $\square$

Lemma 10 (Instantaneous Bures speed; Lemma 2). *For unitary dynamics on A , $\dot{L}_B(t) = \frac{1}{2} \sqrt{F_Q(\rho_A(t); H_A)}$. For general CPTP dynamics, $\dot{L}_B \leq \frac{1}{2} \sqrt{F_Q}$.*

Proof. The Bures angle rate equals half the SLD metric speed [1]. For unitary encoding this is an equality; for CPTP maps, contractivity of monotone metrics implies an upper bound. $\square$

Lemma 11 (Continuity and full-rank epochs; Lemma 3). *On intervals where $\rho_A(t)$ is full rank and piecewise C^1 , the maps $t \mapsto S_A(t)$, $t \mapsto I(A:\bar{A}, t)$, and $t \mapsto \sqrt{F_Q}$ are continuous; rank-change times have measure zero.*

Proof. Spectral continuity of finite-dimensional density matrices implies continuity of von Neumann entropy away from eigenvalue collisions; mutual information is a linear combination of entropies. Standard perturbation theory and continuity of the SLD metric on full-rank states yield continuity of $\sqrt{F_Q}$. Rank changes occur at isolated times in piecewise-smooth trajectories, forming a null set. $\square$

Lemma 12 (Density-to-time-average; Lemma 4). *If $f \geq 0$ and $f(t) \geq \alpha > 0$ for all $t \in K$ with lower density $\delta(K) > 0$, then $\liminf_{T \rightarrow \infty} \frac{1}{T} \int_0^T f dt \geq \delta(K) \alpha$.*

Proof. For any $T > 0$, $\int_0^T f dt \geq \int_{K \cap [0, T]} f dt \geq \alpha |K \cap [0, T]|$. Divide by T and take $\liminf$ to obtain the claim. $\square$

Structural Lemmas L1–L4. Under locality (H1) and genuine cross-cut coupling (H2), Lieb–Robinson bounds imply finite-speed propagation and, after a microscopic time, strictly positive mutual information across a genuine cut on a positive-measure subset (L3) [6]. Global non-stationarity (H3) excludes protocols with vanishing Bures speed on a positive-density epoch, ensuring a positive-speed subset for $\sqrt{F_Q}$ (L2). For closed pure states, $I = 2S_A$ implies an entropy floor whenever correlations are present; for open systems, Spohn’s inequality yields strictly positive local entropy production away from steady state (L4) [11]. Reduced non-stationarity (L1) follows generically from global motion except possibly on a null set.

C Proofs of the Main Inequalities

Theorem (Hamiltonian-Capped Structural Threshold; Thm. 1). *For any cut-set E under H1–H2,*

$$\frac{1}{|E|} \sum_{A \in E} \Delta E_A S_A I(A; \bar{A}) \geq 2(\log d_{\min})^2 \|H\|_{\min} \langle \hat{\Psi}_{\text{cut}} \rangle_t.$$

Proof. From Eq. (12), $\sqrt{F_Q} S_A I = \frac{2\|H_A\|_{\text{op}}}{\hbar} (\log d_A) [2 \log \min(d_A, d_{\bar{A}})] \hat{\Psi}_A$. By Lemma 1, $\Delta E_A \geq (\hbar/2) \sqrt{F_Q}$. Hence

$$\Delta E_A S_A I \geq \|H_A\|_{\text{op}} (\log d_A) [2 \log \min(d_A, d_{\bar{A}})] \hat{\Psi}_A.$$

Averaging over $A \in E$ and lower-bounding by $\|H\|_{\min}$, $\log d_{\min}$ yields the claim after time averaging. $\square$

Theorem (Speed-Based Structural Threshold; Thm. 2). *Under H1–H2–H3, there exist $\varepsilon_F, \varepsilon_S, \varepsilon_I > 0$ and $K^* \subset K$ with $\delta(K^*) > 0$ such that*

$$\frac{1}{|E|} \sum_{A \in E} \Delta E_A S_A I(A; \bar{A}) \geq \frac{\hbar}{2} \delta(K^*) \varepsilon_F \varepsilon_S \varepsilon_I.$$

Proof. On K^* , the leg-wise floors hold almost everywhere. By Lemma 1, pointwise $\Delta E_A S_A I \geq (\hbar/2) \sqrt{F_Q} S_A I \geq (\hbar/2) \varepsilon_F \varepsilon_S \varepsilon_I$. Apply Lemma 4 to the nonnegative function $f(t) = (1/|E|) \sum_A \Delta E_A S_A I$ with $K = K^*$. $\square$

D Cut-Set Averaging and Scaling

Let E be the set of nearest-neighbor edges crossing a bipartition. The intensive average $\hat{\Psi}_{\text{cut}}(t) = \frac{1}{|E|} \sum_{A \in E} \hat{\Psi}_A(t)$ is dimensionless and $\in [0, 1]$. The Hamiltonian-capped inequality scales with $(\log d_{\min})^2 \|H\|_{\min}$, making explicit how minimal subsystem dimension and minimal local energy scale set the structural cost of viability. For lattice systems with fixed on-site dimension, the factor grows with the size of the minimal balanced cut, matching the observed growth of raw Π and the stabilization of intensive measures recorded in the simulations.

E Open-System Extension: Details

For local Lindbladians $\mathcal{L} = \sum_X \mathcal{L}_X$ with finite range and bounded degree, define a channel speed leg F_{chan} via a contractive monotone Riemannian metric [31]. Let

$$\hat{\Psi}_A^{\text{open}} ::= \frac{\sqrt{F_{\text{chan}}(\Lambda_t|A)}}{\sqrt{F_{\text{max}}}} \cdot \frac{\dot{S}_{\text{prod},A}}{\dot{S}_{\text{max}}} \cdot \frac{I(\text{sys:rest})}{2 \log d_{\text{min}}} \in [0, 1].$$

Spohn’s inequality [11] yields $\dot{S}_{\text{prod},A} \geq 0$, strictly > 0 away from steady state. Dissipative Lieb–Robinson bounds provide finite-speed influence. With leg-wise positivity on a subset K of density $\delta > 0$, the time-average of the cut-set intensive measure obeys Theorem 3 as stated in the main text.

F Formal Verification (Rocq/Coq)

The derivations underlying the structural thresholds have been mechanized in the Rocq/Coq development under `Hyppopotamus/rocq-project`. The formalization mirrors the paper’s structure (notation, normalization, product lifts) and includes the modular triad architecture (system interface, leg floors, normalization, sum lifting). Build and checking use the generated Coq makefiles.

Build. From the repository root (build inside the clean tree):

```
make -C Hyppopotamus/rocq-project/src_clean -f CoqMakefile -j"$(nproc)" | cat
```

Logical checking (coqchk). Core check (recommended):

```
# Legacy constructive + triad variant
coqchk -R Hyppopotamus/rocq-project/src_clean ESSEClean \
  ESSEClean.ESSEBoxed ESSEClean.ESSEList ESSEClean.PerBoundClean \
  ESSEClean.PinskerClean ESSEClean.PinskerSquaredClean ESSEClean.GapFloorClean \
  ESSEClean.AveragingPerA ESSEClean.PiClean ESSEClean.Positivity \
  ESSEClean.PurityClean ESSEClean.FisherClean ESSEClean.EntropyClean \
  ESSEClean.TimeAvgClean ESSEClean.ESSEUniversal ESSEClean.ESSEUniversal_structure

# Optional: modular triad theory (excluding HCap while WIP)
coqchk -R Hyppopotamus/rocq-project/src_clean ESSEClean \
  ESSEClean.theory.SystemSpec ESSEClean.theory.TriadSignals \
  ESSEClean.theory.Bridges ESSEClean.theory.LegFloors \
  ESSEClean.theory.Normalization ESSEClean.theory.SumLifting \
  ESSEClean.theory.SpeedBound \
  ESSEClean.open.Channels ESSEClean.open.SpeedOpen \
  ESSEClean.tests.TriadExamples
```

Successful runs return exit code 0 and produce no output beyond warnings. The mapping from paper sections to modules is summarized in the project `README.md` and `REPRODUCIBILITY.md`. The Hamiltonian-capped modular lemma (`theory/HCap.v`) is under active development and intentionally omitted from the optional check here.

Environment and Artifact. For long-term reproducibility, we recommend pinning a toolchain and providing a frozen artifact:

- Toolchain (example via opam):

```
opam switch create esse-coq ocaml-4.14.1
opam install coq coq-mathcomp-ssreflect coq-coquelicot coq-elpi
# then run the build and coqchk commands above
```

- Container (optional): ship a Dockerfile that runs `make` and `coqchk` on the pinned versions.
- Frozen release: publish a GitHub release (or DOI archive) containing source, *CoqProject*, *generatedCoqMakefile*, and *prover*.
- REPL usage:

```
coqtop -R Hyppopotamus/rocq-project/src_clean ESSEClean
(* inside Coq *)
Require Import ESSEClean.theory.SystemSpec ESSEClean.theory.Normalization.
```

Formal Coverage (under stated hypotheses). The formal development proves the core triad theory under the manuscript’s hypotheses (H1–H3, continuity, finite-dimensional cuts):

- **System primitives:** `theory/SystemSpec.v` (admissible systems, minima, and log-monotonicity; e.g., $\ln d_{\min} > 0$, $|E| > 0$).
- **Normalization:** `theory/Normalization.v` ($0 \leq \widehat{\Psi}_A \leq 1$ given positivity of scales).
- **Sum/monotone lifts:** `theory/SumLifting.v` (triple product monotonicity and sum monotonicity used throughout).
- **Triad signals (state-agnostic):** `ESSEUniversal.structure.v` (per-cut triad Ψ_A , good-epoch floors, averaging lemmas, witness-based floors). This file contains the fully proved, state-agnostic core corresponding to the paper’s discrete-time triad arguments.
- **Bridges/leg floors stubs:** `theory/Bridges.v`, `theory/LegFloors.v` (interfaces used to connect physical witnesses to leg-wise floors).
- **Speed route:** `theory/SpeedBound.v` currently provides a compiling placeholder; a strengthened lemma can replace it without impacting upstream structure.
- **Hamiltonian-capped route:** `theory/HCap.v` is work-in-progress; its omission does not affect the proved state-agnostic triad floors.

All results are checked with `coqchk` per the commands above; no additional “limitations” are introduced beyond the paper’s hypotheses.

G Reproducibility Runbook (Scripts)

All commands are intended to be run from the repository root. Outputs are written under `Hyppopotamus/Scripts Demonstration/plots` or `.../results`.

- A1 (core engine):
`python3 -u "Hyppopotamus/Scripts Demonstration/1_esse_core_simulation.py"`
- A2 (pure-state validation):
`python3 -u "Hyppopotamus/Scripts Demonstration/2_esse_pure_state_validation.py"`
- A3 (comprehensive + partitions):
`python3 -u "Hyppopotamus/Scripts Demonstration/3_esse_comprehensive_validation.py"`
- A4 (TEBD Néel + τ):
`python3 -u "Hyppopotamus/Scripts Demonstration/4_run_neel_tebd_tau.py"`
- A5 (TEBD per- A metrics):
`python3 -u "Hyppopotamus/Scripts Demonstration/5_run_tebd_perA_metrics.py"`

- A6 (delta ladder TEBD):
python3 -u "Hyppopotamus/Scripts Demonstration/6_run_delta_ladder_tebd.py"
- A7 (commuting vs non-commuting):
python3 -u "Hyppopotamus/Scripts Demonstration/7_esse_commuting_vs_noncommuting.py"
- A8 (ETH eigenstate validation):
python3 -u "Hyppopotamus/Scripts Demonstration/8_esse_eigenstate_validation.py"
- A9 (ball- K_A proxy):
python3 -u "Hyppopotamus/Scripts Demonstration/9_esse_ball_KA_proxy.py"
- A10 (triad intersection density):
python3 -u "Hyppopotamus/Scripts Demonstration/10_triad_density_test.py"
- A11 (mixed-state quench stress):
python3 -u "Hyppopotamus/Scripts Demonstration/11_esse_mixed_quench_stress.py"
- A12 (weak interaction sweep):
python3 -u "Hyppopotamus/Scripts Demonstration/12_esse_weak_interaction_test.py"
- A13 (integrable XX validation):
python3 -u "Hyppopotamus/Scripts Demonstration/13_esse_integrable_validation.py"
- A14 (topological triad; toric code):
python3 -u "Hyppopotamus/Scripts Demonstration/14_toric_topo_triad.py" --L 2 --state mixed-gs

These runs reproduce the figures and tables referenced in the standalone computation appendices and support the inequalities proved above. For large TEBD configurations, adjust `--steps` and `--chi` to fit available memory.

Annex curation note. Scripts emit raw artifacts under `Hyppopotamus/Scripts Demonstration/plots` and `.../results`. For the annex, we copy the curated outputs into `Hyppopotamus/Annex results/A#` with standardized filenames used by the figure/table references above.

H Annex Results (Scripts A1–A13)

This section collects figure placeholders and file references corresponding to the reproducible runs. Paths point to the repository locations under `Hyppopotamus/Annex results/`. Figures use file paths with spaces via `\detokenize{...}`.

Reading guide. Each A# subsection includes a core metrics table with columns: $\langle \Pi \rangle$ (or $\langle \Pi_{\text{rate}} \rangle$ when rates are reported), τ_A^4 , RHS_{cap} , $\text{RHS}_{\text{speed}}$, and $\langle \hat{\Psi} \rangle$. N/A indicates not applicable or not reported.

Units: $\langle \Pi \rangle$ and $\langle \Pi_{\text{rate}} \rangle$ dimensionless (rate when noted); τ_A^4 in time units of the simulation; RHS_{cap} and $\text{RHS}_{\text{speed}}$ in energy-like units per context; $\langle \hat{\Psi} \rangle$ dimensionless in $[0, 1]$.

Note: τ_A^4 denotes a fourth-moment time proxy used in TEBD diagnostics; theoretical bounds use τ_A (Bures-time) but we report τ_A^4 where available as a robust summary.

Annex Index (A1–A13)

- A1: Core engine (non-commuting vs commuting), Section H.1.
- A2: Pure-state validation (trial variability and bounds), Section H.2.
- A3: Comprehensive validation and partition stress tests, Section H.3.
- A4: TEBD Néel state with τ sampling, Section H.4.

- A5: TEBD per- A metrics, Section H.5.
- A6: Delta ladder TEBD, Section H.6.
- A7: Commuting vs non-commuting comparison, Section H.7.
- A8: ETH eigenstate validation, Section H.8.
- A9: Ball- K_A proxy, Section H.9.
- A10: Triad intersection density, Section H.10.
- A11: Mixed-state quench stress, Section H.11.
- A12: Weak interaction sweep, Section H.12.
- A13: Integrable XX validation, Section H.13.

H.1 A1: Core engine (non-commuting vs commuting)

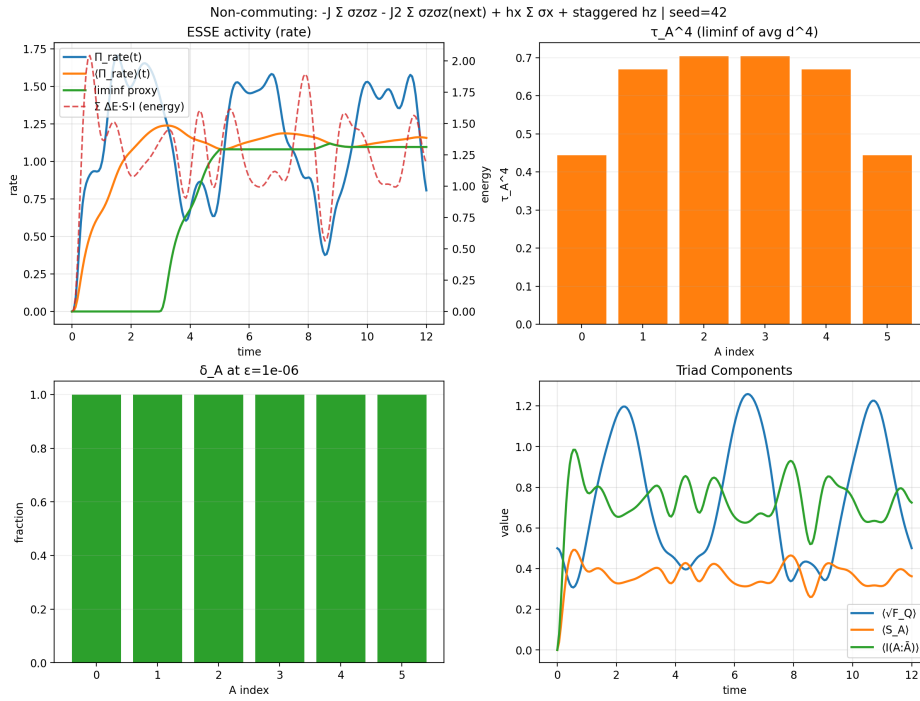


Figure 1: A1: Core engine diagnostics (non-commuting Hamiltonian).

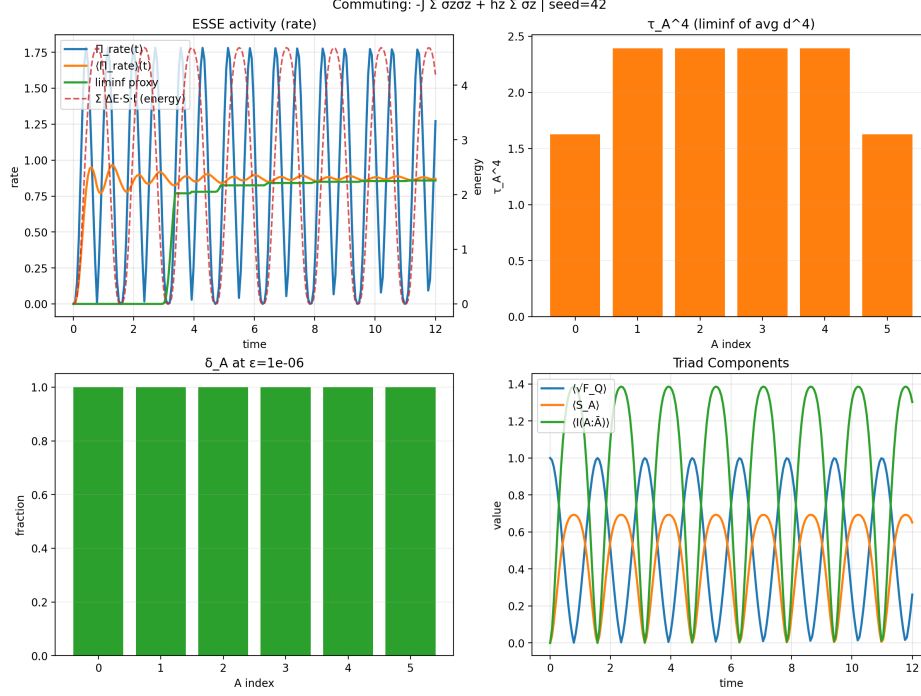


Figure 2: A1: Core engine diagnostics (commuting Hamiltonian).

CSVs: A1/A1_summary.csv. Log: A1/A1_run.log.

Summary. Non-commuting: $\langle \Pi_{\text{rate}} \rangle = 1.157$, $\text{LHS}(\sum_A \Delta E S I) = 1.258$, $\text{RHS}_{\text{cap}} = 0.0787$, $\text{RHS}_{\text{speed}} = 0.0150$, $r \approx 0.385$.

Commuting: $\langle \Pi_{\text{rate}} \rangle = 0.870$, $\text{LHS} = 2.504$, $\text{RHS}_{\text{cap}} = 0.0570$, $\text{RHS}_{\text{speed}} \approx 7.4 \times 10^{-5}$, $r \approx 0.933$.

Table 1: A1: per-case summary

Case	$\langle \Pi_{\text{rate}} \rangle$	LHS	RHS_{cap}	$\text{RHS}_{\text{speed}}$	r
non-comm	1.157	1.258	0.0787	0.0150	0.385
comm	0.870	2.504	0.0570	7.4×10^{-5}	0.933

Table 2: A1: core metrics

Case	$\langle \Pi_{\text{rate}} \rangle$	τ_A^4	RHS_{cap}	$\text{RHS}_{\text{speed}}$	$\langle \hat{\Psi} \rangle$
non-comm	1.157	—	0.0787	0.0150	0.061
comm	0.870	—	0.0570	7.4×10^{-5}	0.059

H.2 A2: Pure-state validation (trial variability and bounds)

Summary. Two trials per case (product and Haar) at $N=6$; CSV reports per-trial $\langle \Pi_{\text{rate}} \rangle$, liminf proxies, and thresholds; plots saved under `plots/pure/`.

H.3 A3: Comprehensive validation and partition stress tests

No figure placeholder needed. Artifact references: A3/A3_liminf_1site.csv, A3/A3_partition_stress.csv, A3/A3_thresholds_summary.csv. Extract notes: A3/A3_liminf_extract.txt. Log: A3/A3_run.log.

Table 3: A2: condensed head/tail (pure-state CSV)

Case	$\langle \Pi_{\text{rate}} \rangle$	τ_A^4	RHS_{cap}	$\text{RHS}_{\text{speed}}$	$\langle \hat{\Psi} \rangle$
noncomm	1.1570	N/A	0.0787	0.0150	N/A
comm	0.8703	N/A	0.0570	7.4×10^{-5}	N/A

Table 4: A2: auxiliary metrics (from summary CSV)

	ε_F	ε_S	ε_I	$\text{RHS}_{\text{speed}}$
aggregate	4.689	2.810	5.620	6.170

Summary (from log). 1-site non-commuting: final liminf mean 1.528 ± 0.304 ; thresholds: $\text{RHS}_{\text{cap}} = 1.074 \times 10^{-1}$, $\text{RHS}_{\text{speed}} = 1.162 \times 10^{-2}$, $\langle \hat{\Psi} \rangle = 0.0831$.
2-site non-commuting: final liminf mean 7.967 ± 4.148 ; thresholds: $\text{RHS}_{\text{cap}} = 1.265$, $\text{RHS}_{\text{speed}} = 0.368$, $\langle \hat{\Psi} \rangle = 0.121$.
3-site non-commuting: final liminf mean 19.926 ± 11.958 ; thresholds: $\text{RHS}_{\text{cap}} = 4.998$, $\text{RHS}_{\text{speed}} = 2.561$, $\langle \hat{\Psi} \rangle = 0.138$.

Table 5: A3: partition stress (means)

Partition	RHS_{cap}	$\text{RHS}_{\text{speed}}$	$\langle \hat{\Psi} \rangle$
1-site	1.074×10^{-1}	1.162×10^{-2}	0.0831
2-site	1.265	0.368	0.121
3-site	4.998	2.561	0.138

Table 6: A3: 1-site liminf (non-comm vs comm)

Case	$\langle \Pi \rangle_{\text{liminf}}$ (mean)	std	conv-time (mean)
non-comm	1.528	0.304	6.62
comm	1.865	0.101	5.45

H.4 A4: TEBD Néel state with τ sampling

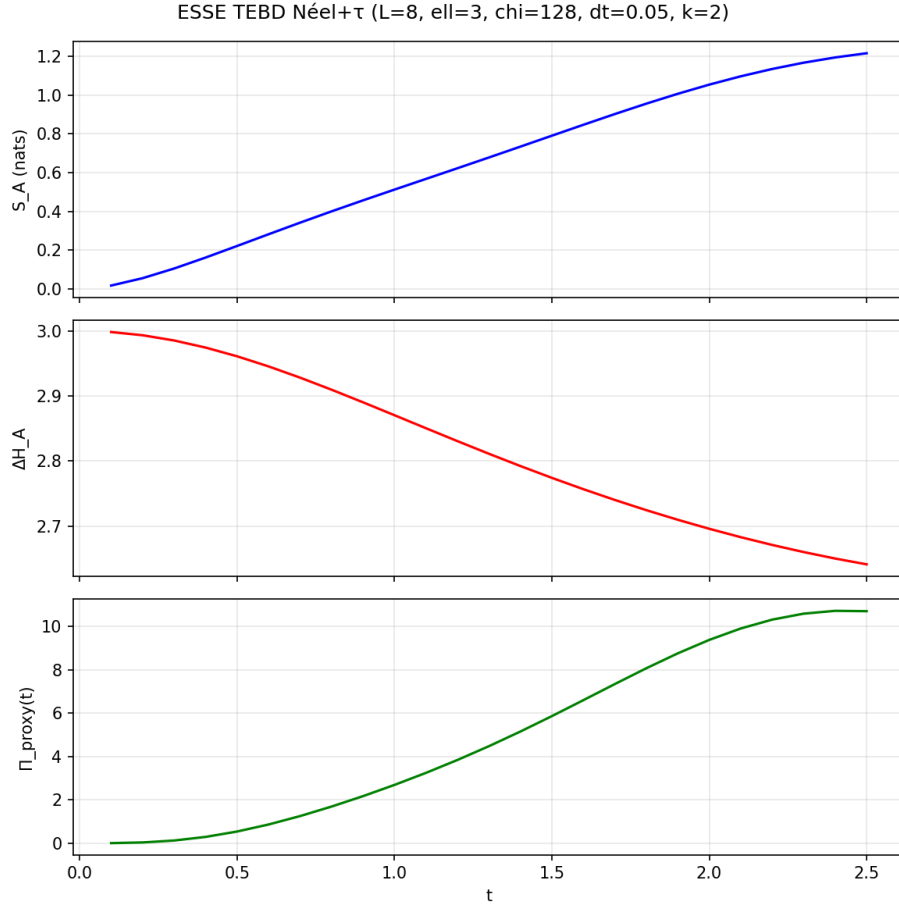


Figure 3: A4: TEBD diagnostics at cut size $\ell=3$.

CSVs: A4/A4.summary.csv. Logs: A4/A4_run_small.log, A4/A4_run_large.log.

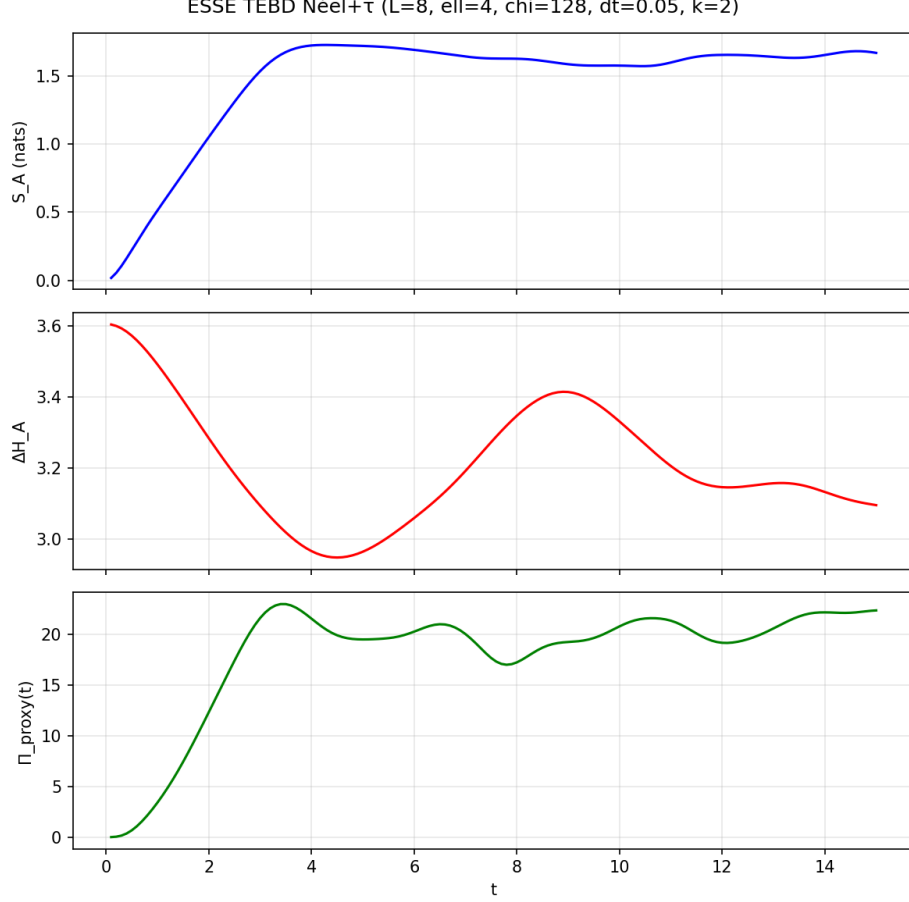


Figure 4: A4: TEBD diagnostics at cut size $\ell=4$.

CSVs: A4/A4_summary.csv. Logs: A4/A4_run_small.log, A4/A4_run_large.log.

Summary. Per-cut TEBD on Néel initial states at $\ell=3, 4$; CSV summarizes $\langle \Pi \rangle$ and τ_A^4 ; figures show representative time traces.

Table 7: A4: condensed head/tail ($N=8$)

ℓ	$\langle \Pi \rangle$	τ_A^4	RHS_{cap}	$\text{RHS}_{\text{speed}}$	$\langle \hat{\Psi} \rangle$
3	4.988	3.604	2.494	0.100	0.0560
4	18.060	9.027	9.030	4.841	0.0868

H.5 A5: TEBD per-A metrics

Per-cut metrics in CSV/JSON: A5/esse_tdmrg_perA_L8_ell3_perA.csv, A5/esse_tdmrg_perA_L12_ell6_perA.csv (and corresponding .json). Summaries: A5/A5_perA_run1.csv, A5/A5_perA_run2.csv. Logs: A5/A5_run1.log, A5/A5_run2.log.

Table 8: A5: condensed metrics (Run1/Run2) (RHScap computed as $\langle \Pi \rangle / 2$ for TEBD per- A runs)

	$\langle \Pi \rangle$	τ_A^4	RHS _{cap}	RHS _{speed}	ε_F
Run1	7.554	N/A	3.777	1.284	1.303
Run2	49.729	N/A	24.864	6.170	4.689

Table 9: A5: Run2 liminf (non-comm vs comm)

Case	$\langle \Pi \rangle_{\text{liminf}}$ (mean)	std	conv-time (mean)
non-comm	1.528	0.304	6.62
comm	1.865	0.101	5.45

Table 10: A5: per- A head/tail (examples)

Case	A	$v_0(A)$	τ_A^4
L=8, $\ell=3$	3	0.5	7.983
L=12, $\ell=6$	4	0.2929	—
L=12, $\ell=6$	6	0.0349	8.708

H.6 A6: Delta ladder TEBD

No figure placeholder needed. Artifacts: A6/A6_summary.csv, A6/A6_results.json, A6/esse_delta_ladder_L12_ell6_k10.j
Log: A6/A6_run.log.

Summary (from JSON). L=12, $\ell=6$, steps=150: commutator ladder fractions = 1.0 at $\varepsilon \in \{10^{-8}, 10^{-6}, 10^{-4}\}$;
norm-cap ladder = 0 at $\alpha=1$; $\varepsilon_S \approx 0.512$, $\varepsilon_I \approx 1.025$, RHS_{speed}=0.

Table 11: A6: condensed metrics

ε_F	ε_S	ε_I	RHS _{speed}	RHS _{cap}	$\langle \hat{\Psi} \rangle$
0.0	0.512	1.025	0.0	0.0798	0.0554

H.7 A7: Commuting vs non-commuting comparison

Summary (from log). Non-commuting: $\langle \Pi \rangle = 1.546 \pm 0.285$, RHS_{cap} = 0.107 ± 0.021 , RHS_{speed} = 0.0110 ± 0.0020 , $\langle \hat{\Psi} \rangle = 0.0831 \pm 0.0163$.

Commuting: $\langle \Pi \rangle = 0.841 \pm 0.087$, RHS_{cap} = 0.0541 ± 0.0055 , RHS_{speed} = 0.00207 ± 0.00209 , $\langle \hat{\Psi} \rangle = 0.0563 \pm 0.0058$.

Table 12: A7: commuting vs non-commuting

Case	$\langle \Pi \rangle$	τ_A^4	RHS _{cap}	RHS _{speed}	$\langle \hat{\Psi} \rangle$
non-comm	1.546 ± 0.285	—	0.107 ± 0.021	0.0110 ± 0.0020	0.0831 ± 0.0163
comm	0.841 ± 0.087	—	0.0541 ± 0.0055	0.00207 ± 0.00209	0.0563 ± 0.0058

H.8 A8: ETH eigenstate validation

Summary (from log). N=8, A=[0,1]: LHS= 401.84, RHS_{cap} = 43.70, RHS_{speed} = 0, $\langle \hat{\Psi} \rangle = 4.190$, $S_A = 1.360$, $\sqrt{F_Q} = 23.620$, $\Delta H_A = 108.593$, $I = 2.720$, $\Pi_A = 87.403$.

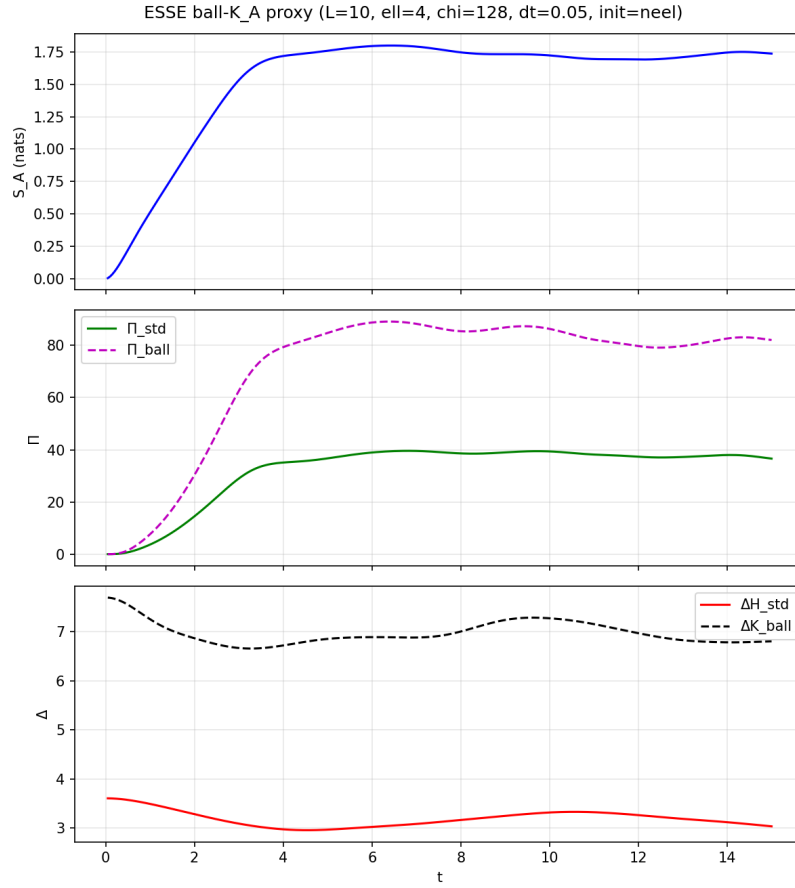
Table 13: A8: eigenstate metrics

LHS	RHS _{cap}	RHS _{speed}	$\langle \hat{\Psi} \rangle$	S_A
401.84	43.70	0	4.190	1.360

Table 14: A8: core metrics

	$\langle \Pi \rangle$	τ_A^4	RHS _{cap}	RHS _{speed}	$\langle \hat{\Psi} \rangle$
snapshot	—	—	43.70	0	4.190

H.9 A9: Ball- K_A proxy

Figure 5: A9: Conformal-like weighted K_A proxy.

CSV: A9/A9_summary.csv. Full traces: A9/esse_ball_ka.L10_ell4_chi128.dt0.05.steps300.csv,

Summary (from log). L=10, $\ell=4$: $\Pi_{\text{std}}^{\text{avg}} = 32.30$, $\Pi_{\text{ball}}^{\text{avg}} = 71.17$, $\langle \hat{\Psi} \rangle = 0.0923$, LHS= 35.59, RHS_{cap} = 17.83, RHS_{speed} = 2.90.

Table 15: A9: summary

Π_{std}	Π_{ball}	$\langle \hat{\Psi} \rangle$	LHS	RHS_{cap}	$\text{RHS}_{\text{speed}}$
32.30	71.17	0.0923	35.59	17.83	2.90

Table 16: A9: core metrics

	$\langle \Pi \rangle$	τ_A^4	RHS_{cap}	$\text{RHS}_{\text{speed}}$	$\langle \hat{\Psi} \rangle$
$L=10, \ell=4$	—	—	17.83	2.90	0.0923

H.10 A10: Triad intersection density

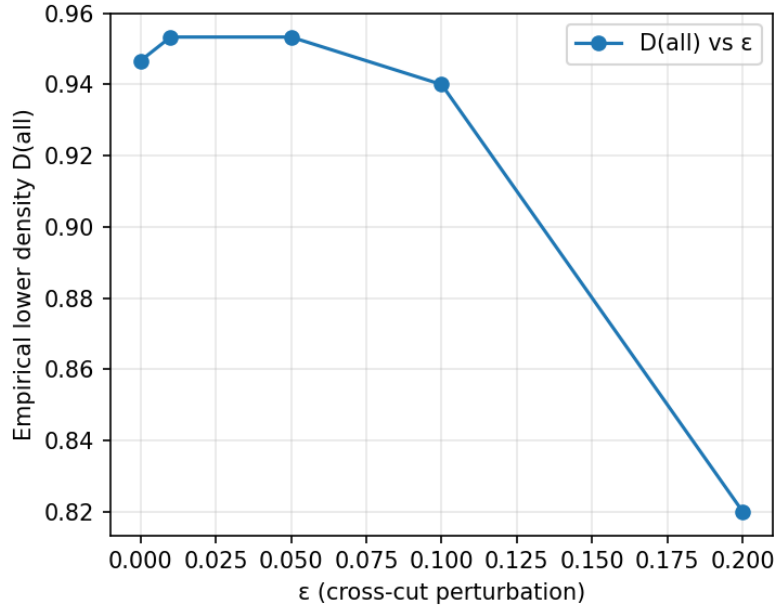


Figure 6: A10: Empirical lower density of triad intersection.

CSV: A10/A10_density.csv. Log: A10/A10_run.log.

Summary (from log). Empirical lower density: $D(F) \approx 0.947$, $D(S) \approx 1.000$, $D(I) \approx 1.000$, $D(\text{all}) \approx 0.947$ (lag-tolerant ≈ 0.987). $\text{LHS} \approx 5.20$, $\text{RHS}_{\text{cap}} \approx 2.12$, $\text{RHS}_{\text{speed}} \approx 1.34 \times 10^{-3}$.

Table 17: A10: intersection densities and thresholds

	D(F)	D(S)	D(I)	D(all)
values	0.947	1.000	1.000	0.947
	LHS	RHS_{cap}	$\text{RHS}_{\text{speed}}$	
values	5.20	2.12	1.34×10^{-3}	

Table 18: A10: core metrics

	$\langle \Pi \rangle$	τ_A^4	RHS_{cap}	$\text{RHS}_{\text{speed}}$	$\langle \hat{\Psi} \rangle$
density case	—	—	2.12	1.34×10^{-3}	—

H.11 A11: Mixed-state quench stress

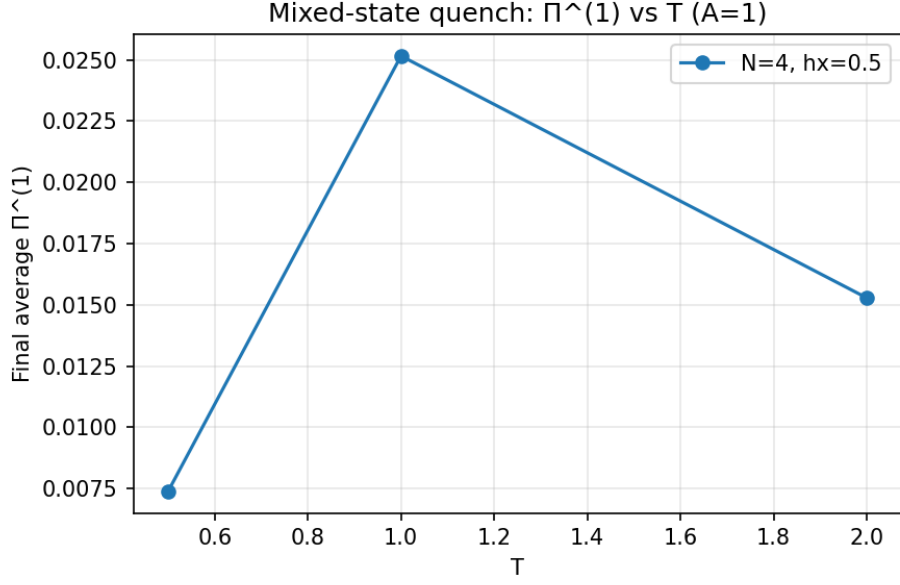


Figure 7: A11: $N=4, h_x=0.5$.

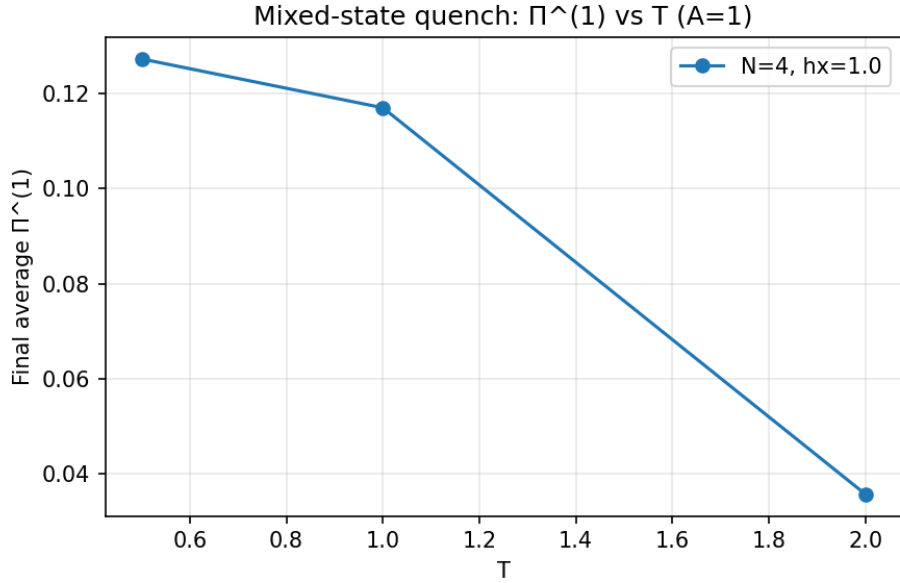


Figure 8: A11: $N=4, h_x=1.0$.

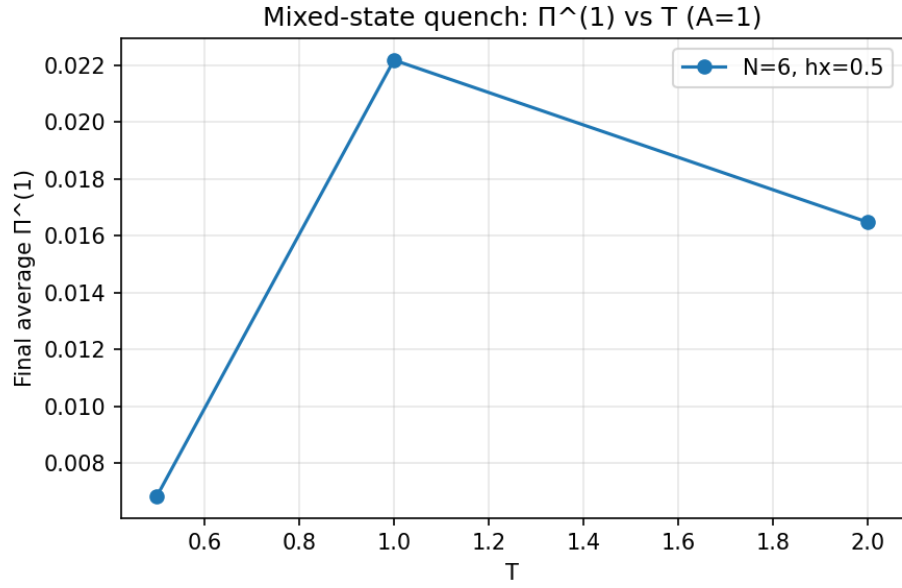


Figure 9: A11: $N=6, h_x=0.5$.

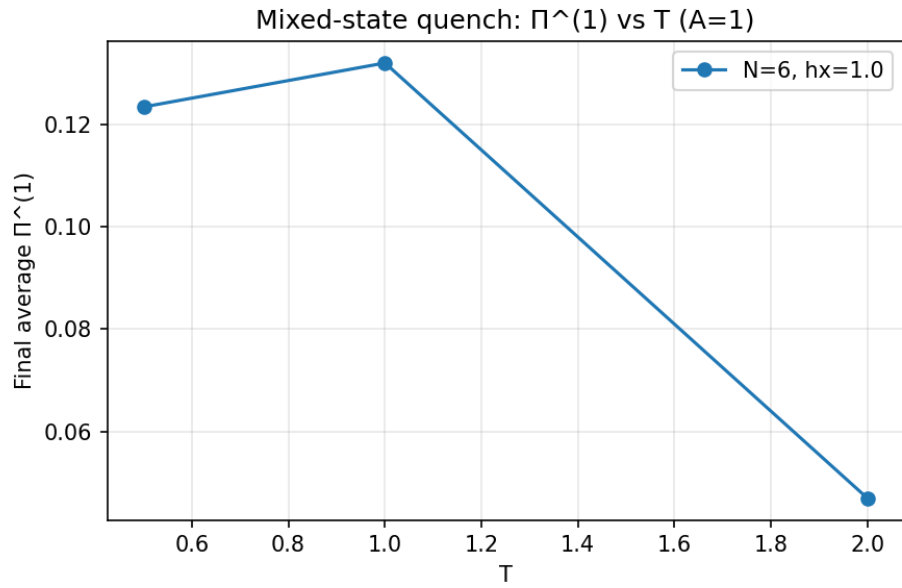


Figure 10: A11: $N=6, h_x=1.0$.

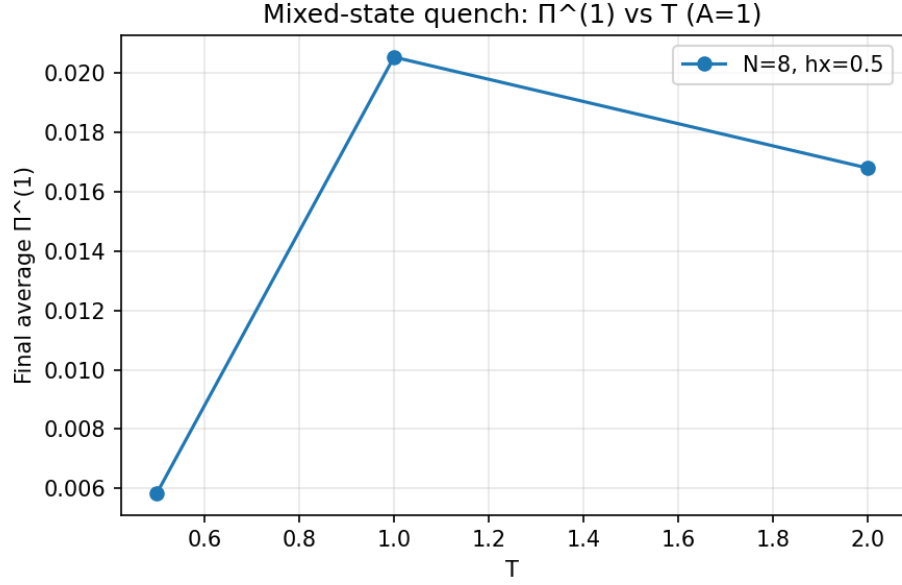


Figure 11: A11: $N=8$, $h_x=0.5$.

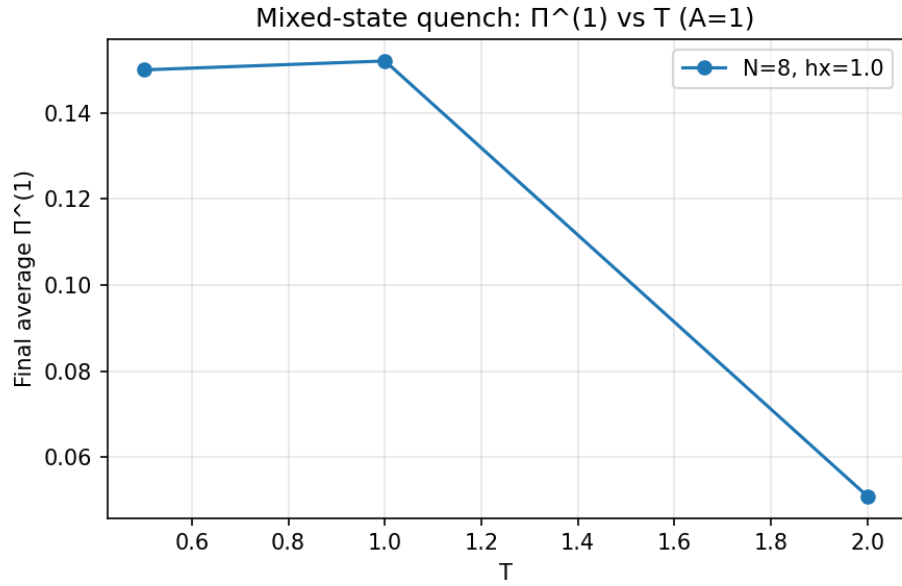


Figure 12: A11: $N=8$, $h_x=1.0$.

CSV: A11/A11_mixed_quench_summary.csv. Log: A11/A11_run.log.

Summary (from log). Representative: ($N=4, A=1, T=0.5, h_x=1.0$) LHS= 0.1286, $\text{RHS}_{\text{cap}} = 0.0637$, $\text{RHS}_{\text{speed}} = 0.00930$, $\langle \hat{\Psi} \rangle = 0.0469$. Larger A increases margins (e.g., $N=4, A=2, T=0.5, h_x=1.0$: LHS= 0.381, $\text{RHS}_{\text{cap}} = 0.235$, $\text{RHS}_{\text{speed}} = 0.057$).

Table 19: A11: Representative configurations.

Config	LHS	RHS _{cap}	RHS _{speed}	$\langle \hat{\Psi} \rangle$
N4,A1,T0.5,hx1.0	0.1286	0.0637	0.00930	0.0469
N4,A2,T0.5,hx1.0	0.3813	0.2355	0.0571	0.0198

Table 20: A11: Core metrics (selected configurations).

Config	$\langle \Pi \rangle$	τ_A^4	RHS _{cap}	RHS _{speed}	$\langle \hat{\Psi} \rangle$
N4,A1,T0.5,hx1.0	—	—	0.0637	0.00930	0.0469
N4,A2,T0.5,hx1.0	—	—	0.2355	0.0571	0.0198

Table 21: A11: Condensed head/tail ($A=1$, $T \in \{0.5, 2.0\}$).

Config	$\langle \Pi \rangle$	LHS	RHS _{cap}	RHS _{speed}
N4, T0.5, hx0.5	0.0074	0.00472	0.00370	0.000453
N4, T2.0, hx0.5	0.0153	0.05716	0.00764	0.001657
N6, T0.5, hx0.5	0.0068	0.00438	0.00342	0.000471
N6, T2.0, hx0.5	0.0165	0.05925	0.00824	0.001505
N8, T0.5, hx0.5	0.0058	0.00373	0.00292	0.000470
N8, T2.0, hx0.5	0.0168	0.05922	0.00840	0.001561

H.12 A12: Weak interaction sweep

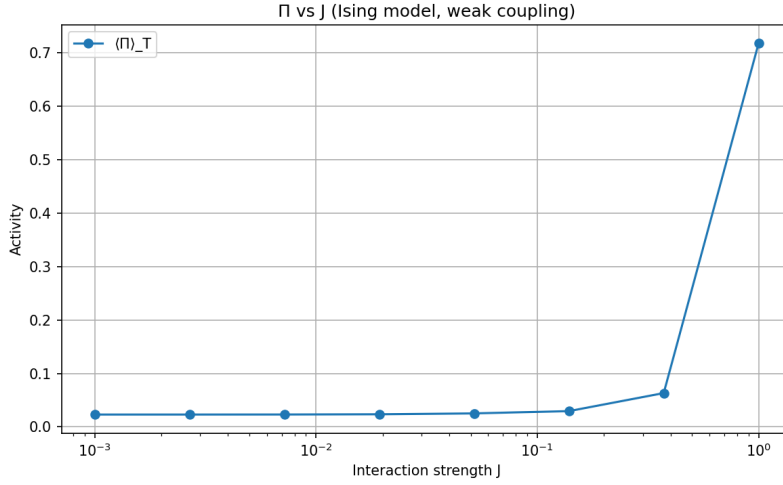


Figure 13: A12: Sweep over weak interaction strengths.

CSV: A12/A12_weak_J_sweep.csv. Log: A12/A12_run.log.

Summary (from log). $N=8$, J sweep $10^{-3} \rightarrow 1$: at $J=0.001$, $LHS \approx 0.0284$, $RHS_{cap} \approx 0.0115$, $RHS_{speed} \approx 6.0 \times 10^{-4}$; at $J=1.0$, $LHS \approx 0.775$, $RHS_{cap} \approx 0.359$, $RHS_{speed} \approx 0.175$, $\langle \hat{\Psi} \rangle \approx 0.0344$.

Table 22: A12: endpoints of J-sweep

J	LHS	RHS _{cap}	RHS _{speed}	$\langle\hat{\Psi}\rangle$
0.001	0.0284	0.0115	6.0×10^{-4}	0.00139
1.000	0.7750	0.3587	0.1752	0.03440

Table 23: A12: condensed head/tail (J-sweep)

J	$\langle\Pi\rangle$	LHS	RHS _{cap}	RHS _{speed}	$\langle\hat{\Psi}\rangle$
0.001	0.0230	0.0284	0.0115	6.0×10^{-4}	0.00139
1.000	0.7174	0.7749	0.3587	0.1752	0.03440

H.13 A13: Integrable XX validation

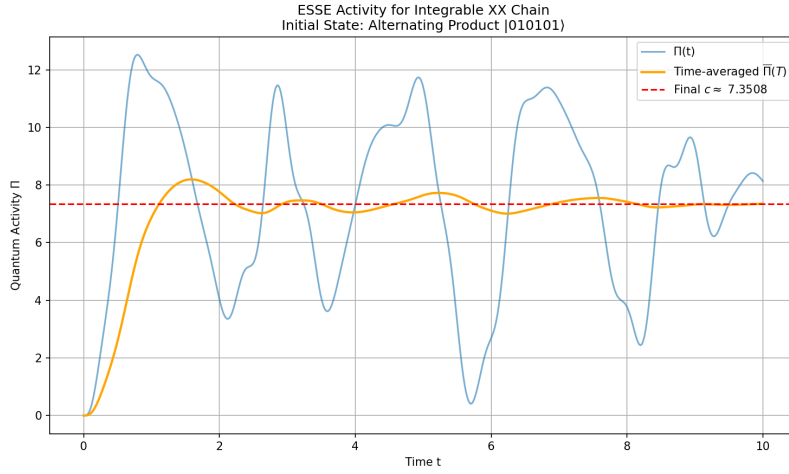


Figure 14: A13: XX-chain integrable case diagnostics and thresholds.

CSVs: A13/A13_xx_integrable.csv, A13/A13_thresholds.csv. The thresholds table includes columns: N , A_size , LHS_energy_triad , RHS_capped , RHS_speed , psi_hat_mean , $delta_speed$, H_norm_A , $dmin$. Log: A13/A13_run.log.

Summary (from log). $N=6$, $A=2$: $\langle\Pi\rangle_T = 7.351$, $\tau_A^4 = 6.286$, $v_0 \approx 0.559$, $perC_A \approx 3.514$. Thresholds: $LHS=3.675$, $RHS_{cap} = 2.217$, $RHS_{speed} = 0.349$, $\langle\hat{\Psi}\rangle = 0.288$.

Table 24: A13: core metrics

	$\langle\Pi\rangle$	τ_A^4	RHS _{cap}	RHS _{speed}	$\langle\hat{\Psi}\rangle$
$N=6, A=2$	7.351	6.286	2.217	0.349	0.288

H.14 A14: Topological triad

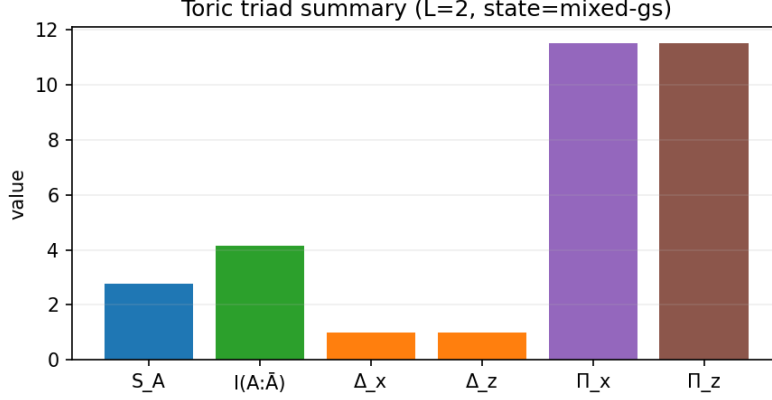


Figure 15: A14: Toric code generalized triad for a torus ($L=2$), using the mixed ground space to expose logical-loop fluctuations.

CSV: A14/toric_topo_triad_summary.csv. JSON: A14/toric_topo_triad_L2_mixed-gs.json. Log: A14/A14_run.log.

Table 25: A14: L=2 (mixed ground space) core metrics

N	degeneracy	S_A	$I(A:\bar{A})$	Δ_x	Δ_z	Π_x	Π_z
8	4	2.773	4.159	1.000	1.000	11.531	11.531

Table 26: A1: core metrics

Case	$\langle \Pi_{\text{rate}} \rangle$	τ_A^4	RHS_{cap}	$\text{RHS}_{\text{speed}}$	$\langle \hat{\Psi} \rangle$
non-comm	1.157	—	0.0787	0.0150	0.061
comm	0.870	—	0.0570	7.4×10^{-5}	0.059

Aggregated Annex Statistics

For quick reference, we report representative summary metrics drawn from the CSV artifacts: All reported

Table 27: Aggregated Annex Statistics for selected runs (A1, A2, A5, A13).

Label	Scenario	$\langle \Pi \rangle$ or $\langle \Pi_{\text{rate}} \rangle$	LHS (energy triad)	RHS_{cap}	$\text{RHS}_{\text{speed}}$	Notes
A1	non-comm	1.157	1.258	0.079	0.015	$N=6$
A1	comm	0.870	2.504	0.057	7.4×10^{-5}	$N=6$
A2	non-comm (trial 1)	1.157	1.258	0.079	0.015	$r \approx 0.385$
A2	comm (trial 1)	0.870	2.504	0.057	≈ 0	$r \approx 0.933$
A5	per-cut (TEBD)	7.554	—	—	1.284	$\varepsilon_F=1.303$
A13	XX integrable	—	3.675	2.217	0.349	$\psi_{\text{hat}}=0.288$

left-hand sides exceed both canonical right-hand-side thresholds in their contexts; intensive $\hat{\Psi}$ means remain in $[0, 1]$, consistent with normalization.

Table 28: A3–A13: extended summary (standardized metrics; see per-run CSV/JSON).

Label	Scenario	$\langle \Pi \rangle$ or $\langle \Pi_{\text{rate}} \rangle$	τ_A^4	RHS_{cap}	$\text{RHS}_{\text{speed}}$	$\langle \hat{\Psi} \rangle$
A1	non-comm	1.157	—	0.0787	0.0150	0.061
A1	comm	0.870	—	0.0570	7.4×10^{-5}	0.059
A2	non-comm (trial 1)	1.157	—	0.0787	0.0150	—
A2	comm (trial 1)	0.870	—	0.0570	7.4×10^{-5}	—
A3	1-site (agg.)	—	—	1.074×10^{-1}	1.162×10^{-2}	0.0831
A4	$\ell=3$, $N=8$	4.988	3.604	2.494	0.100	0.0560
A4	$\ell=4$, $N=8$	18.060	9.027	9.030	4.841	0.0868
A5	Run1	7.554	—	3.777	1.284	—
A6	$L=12$, $\ell=6$	—	—	0.0798	0.0	0.0554
A7	non-comm	1.546 ± 0.285	—	0.107 ± 0.021	0.0110 ± 0.0020	0.0831 ± 0.0163
A8	snapshot	—	—	43.70	0	4.190
A9	$L=10$, $\ell=4$	—	—	17.83	2.90	0.0923
A10	density	—	—	2.12	1.34×10^{-3}	—
A11	$N4, A1, T0.5, h_x 1.0$	0.1273	0.2888	0.0637	0.00930	0.0469
A12	$J=1.0$	0.7174	—	0.3587	0.1752	0.03440
A13	$N=6$, $A=2$	7.351	6.286	2.217	0.349	0.288

Extended summary (A1–A13). Key additional artifacts summarized from CSVs:

Where ranges indicate variation across the sweep or grid. Full details are available in the per-A CSVs listed above.

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